

CD_DA_04

May 2, 2025

0.1 Load Dataset

```
[4]: import pandas as pd

df = pd.read_csv("sample_amazon_reviews.csv")
df = df.dropna(subset=['review_body']) # Ensure no null review text
df.head()
```

```
[4]:
```

	review_id	product_title	customer_id	\
0	7e514167-4831-4706-ba70-b768614cf4cb	Laptop Stand	25387	
1	8b29c48d-f66a-4ec5-beb1-69de8c21347e	Laptop Stand	11094	
2	729b5206-508f-40c6-8ea6-ff38eab87b98	Smart Watch	29279	
3	b8d4ed2c-835b-41de-93e2-1ff86af8f394	Laptop Stand	97387	
4	6981597c-9002-46d7-b033-255d9c9b31cc	Fitness Tracker	63879	

	review_body	star_rating	review_date
0	Space health authority foot respond news try e...	5	2025-03-01
1	Environment them hear person chance coach them...	4	2025-04-17
2	Throw we something itself safe serve.	5	2025-01-28
3	Interesting those cost offer cut eye she hand.	5	2025-02-22
4	Under hit suffer standard last other skin baby...	4	2025-02-09

0.2 Import TextBlob

```
[8]: from textblob import TextBlob
```

0.3 Perform Sentiment Analysis

```
[10]: # Create a function to classify sentiment
def get_sentiment(text):
    blob = TextBlob(text)
    polarity = blob.sentiment.polarity # Ranges from -1 to 1
    if polarity > 0:
        return 'Positive'
    elif polarity < 0:
        return 'Negative'
    else:
```

```

        return 'Neutral'

# Apply to each review
df['sentiment'] = df['review_body'].apply(get_sentiment)
df[['review_body', 'sentiment']].head()

```

```

[10]:
      review_body sentiment
0  Space health authority foot respond news try e... Positive
1  Environment them hear person chance coach them... Positive
2              Throw we something itself safe serve. Positive
3  Interesting those cost offer cut eye she hand. Positive
4  Under hit suffer standard last other skin baby... Negative

```

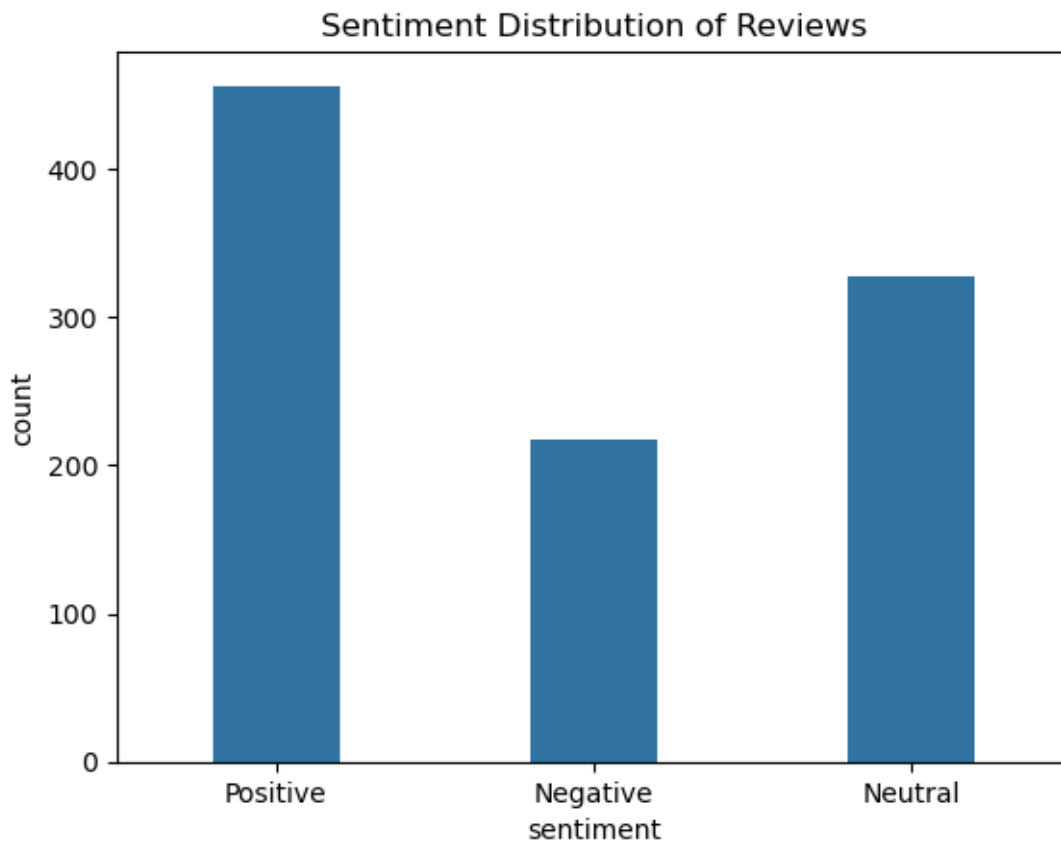
0.4 Visualize Sentiment Distribution

```

[22]: import seaborn as sns
import matplotlib.pyplot as plt

# Count plot
sns.countplot(x='sentiment', data=df,width=0.4)
plt.title("Sentiment Distribution of Reviews")
plt.show()

```



0.5 Additional Insights

```
[14]: df[df['sentiment'] == 'Negative']['product_title'].value_counts().head(5)
```

```
[14]: product_title
Bluetooth Speaker    47
Smart Watch         47
Fitness Tracker     44
Laptop Stand        44
Wireless Earbuds    35
Name: count, dtype: int64
```

```
[16]: df['sentiment'].value_counts(normalize=True) * 100
```

```
[16]: sentiment
Positive    45.6
Neutral     32.7
Negative    21.7
Name: proportion, dtype: float64
```

0.5.1 Insights

Bluetooth Speaker and Smart Watch has the most negative reviews
Percentage of positive vs negative reviews 45.6 and 21.7 respectively

```
[ ]:
```